



Figure 5: Big Data market forecast (Image credits: Google Images)

long-term conditions of patients by using in-home monitoring devices, which measure vital signs and check the progress of patients to improve their health and reduce both hospital admissions and visits to doctors' clinics.

3. Manufacturing companies also deploy sensors in their products to gather data remotely, as in the case of General Motor's OnStar or Renault's R-Link. These help in delivering communications, navigation and security services. They also reveal usage patterns, rates of failure and other such opportunities for product improvement that can further reduce assembly and development costs.
4. The phenomenal increase in the use of smartphones and other GPS devices provides advertisers an opportunity to target their consumers when they are in close proximity to a store, restaurant or a coffee shop. Retailers know the avid buyers of their products better. The use of various social media and Web log files from their e-commerce sites helps them get information about those who didn't buy their products and also the reason for why they chose not to. This can lead to more effective micro, customer-targeted marketing campaigns as well as improved supply chain efficiencies, as a result of more accurate demand planning.
5. Finally, different social media websites like Facebook, Instagram, Twitter and LinkedIn wouldn't have existed without Big Data. The personalised experience provided by them to their different users can only be delivered by storing and using all the available data about that user or member.

How secure is Big Data for enterprise apps?

As it plays around with all sorts of significant data belonging to several organisations which may or may not be related to each other, or their users, it is very important that Big Data should have a high grade of security so that there is no fear among the several enterprises implementing it. Big Data basically provides a comprehensive data security approach.

1. It ensures that the right people (internal or external) get access to the appropriate information and data at the right time and at the right place, through the right channel (typically using Kerberos).

2. High security prevents malicious attacks and it also protects the information assets of the organisation by encrypting (using Cloudera Navigator Encrypt) and securing the data while it is in motion or at rest.
3. It also enables all organisations to separate their different roles and responsibilities, and protect all sensitive data without compromising on the privileged user access like administration of DBAs, etc, using various data masking and subset techniques.
4. It also extends auditing, monitoring and compliance reporting across all traditional data management to the big data systems.

Surprising statistics

- Ninety per cent of the world's data has been created in the last two years, and Big Data is projected to grow into a US\$ 50 billion market by 2017, up from US\$ 12 billion in 2012.
- Seventy per cent of the digital universe (900 exabytes) is being generated by users.
- Enterprises store 80 per cent of all their data.
- The White House administration is investing US\$ 200 million in Big Data research projects.
- China will account for one-fifth of the world's data by 2020.
- A 10 per cent increase in data accessibility translates into an additional US\$ 65.7 billion in net income for a typical Fortune 1000 company.

According to Tom Peters, a best selling author of business management books, "Organisations that do not understand the overwhelming importance of managing data and information as tangible assets in the new economy, will not survive."

Hence, the Big Data promise has really motivated businesses and enterprises to invest. We need to continue researching and designing new technologies and techniques to provide a fast and reliable path for various businesses to adopt Big Data. We need to keep improving our skills and managing various open source tools and technologies to leverage the best Big Data capabilities. This will help various enterprises to manage the increasing data sets and use them effectively. 

References

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